

Text-Guided Brain Tumor Segmentation using Vision Language Models with FLAIR MRI and Radiology Reports

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Abstract

Brain tumor segmentation is a critical task in medical image analysis for accurate diagnosis, treatment planning, and monitoring. Traditional segmentation methods primarily rely on imaging modalities alone and often fail to utilize valuable semantic information available in radiology reports. In this work, we propose a multimodal deep learning framework that integrates FLAIR MRI images from the BraTS 2020 dataset with textual radiology descriptions from the TextBraTS dataset using a Vision Language Model (VLM)-based architecture.

The proposed system employs a multimodal 3D U-Net architecture incorporating a 3D image encoder, CLIP-based text encoder, multimodal fusion module, and segmentation decoder. The framework learns aligned latent representations between MRI scans and corresponding textual descriptions to improve tumor localization and segmentation accuracy.

Experimental results demonstrate strong segmentation performance with a validation Dice Score of 0.9659 and IoU of 0.9347. Additional analyses including Grad-CAM visualizations, t-SNE embedding analysis, and error analysis further validate the effectiveness of the proposed multimodal framework.

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1 Introduction

Brain tumors are among the most life-threatening neurological disorders, requiring precise localization and segmentation for effective treatment planning. Magnetic Resonance Imaging (MRI) is widely used for brain tumor diagnosis due to its superior soft tissue contrast. However, relying solely on imaging information may limit segmentation performance because radiological observations and semantic descriptions are not explicitly incorporated.

Recent advances in Vision Language Models (VLMs) have enabled multimodal learning by jointly processing visual and textual data. Inspired by these developments, this project integrates FLAIR MRI scans and radiology-style textual descriptions to perform text-guided brain tumor segmentation.

The proposed framework utilizes:

- BraTS 2020 FLAIR MRI dataset
- TextBraTS radiology reports
- CLIP-based text encoder
- 3D U-Net segmentation backbone
- Multimodal fusion mechanism

The system demonstrates how textual context improves segmentation performance by enhancing feature representation learning.

2 Objective

The primary objectives of this project are:

- Develop a multimodal segmentation framework combining MRI images and textual reports.
- Learn joint image-text representations using Vision Language Models.
- Improve segmentation performance using textual guidance.
- Evaluate the effectiveness of multimodal learning through quantitative and qualitative analysis.

3 Dataset Description

3.1 BraTS 2020 FLAIR MRI Dataset

The BraTS 2020 dataset contains preprocessed brain MRI scans and corresponding tumor segmentation masks.

Dataset Characteristics:

- MRI modality used: FLAIR

- Input size: $128 \times 128 \times 128$
- Segmentation masks contain multiple tumor regions
- Stored in NumPy (.npy) format

3.2 TextBraTS Dataset

The TextBraTS dataset contains radiology-style textual descriptions corresponding to each patient.

Text Information Includes:

- Tumor location
- Edema description
- Structural abnormalities
- Signal intensity patterns

Each patient has:

- 155 MRI slices
- One corresponding textual report

The project aligns patient-specific MRI scans with their associated textual descriptions.

4 Methodology

4.1 Overall Architecture

The proposed architecture consists of four major components:

1. 3D Image Encoder
2. CLIP-based Text Encoder
3. Multimodal Fusion Module
4. 3D Segmentation Decoder

4.2 Image Encoder

A 3D U-Net style encoder extracts hierarchical spatial features from volumetric MRI scans.

Feature Extraction Levels:

- Level 1: 32 channels
- Level 2: 64 channels
- Level 3: 128 channels
- Level 4: 256 channels
- Bottleneck: 512 channels

4.3 Text Encoder

The textual descriptions are processed using the CLIP text encoder:

- Tokenization using CLIP tokenizer
- Text embedding generation
- Semantic feature extraction

4.4 Multimodal Fusion

Image and text features are fused at the bottleneck stage using a multimodal fusion mechanism.

The fusion module enables:

- Semantic alignment
- Cross-modal feature learning
- Text-guided segmentation

4.5 Decoder

The decoder reconstructs segmentation masks using:

- Upsampling operations
- Skip connections
- Convolutional refinement blocks

Final output size:

$$4 \times 128 \times 128 \times 128$$

representing multi-class tumor segmentation.

5 Implementation Details

5.1 Programming Framework

- Language: Python
- Deep Learning Framework: PyTorch
- Transformer Library: HuggingFace Transformers
- Visualization: Matplotlib

5.2 Project Structure

Brain Tumor/

data/
dataset/
models/
training/
inference/
visualization/
outputs/
utils/

5.3 Training Configuration

Parameter	Value
Batch Size	1
Learning Rate	$1e^{-4}$
Optimizer	AdamW
Loss Function	Dice + BCE Loss
Epochs	50
Early Stopping Patience	5
MRI Input Size	$128 \times 128 \times 128$

Table 1: Training Configuration

6 Experimental Results

6.1 Best Model Performance

The proposed multimodal segmentation framework achieved strong performance on the validation dataset.

Metric	Value
Validation Dice Score	0.9659
Validation IoU	0.9347
Lowest Validation Loss	0.1012
Best Epoch	5

Table 2: Best Model Performance

6.2 Final Error Analysis

Metric	Value
Mean Dice Score	0.9625
Minimum Dice Score	0.9055
Maximum Dice Score	0.9930
Standard Deviation	0.0194
Failure Cases (≤ 0.90 Dice)	0

Table 3: Error Analysis Summary

The model demonstrated highly stable segmentation performance with no severe failure cases.

7 Training Performance

7.1 Loss vs Epoch

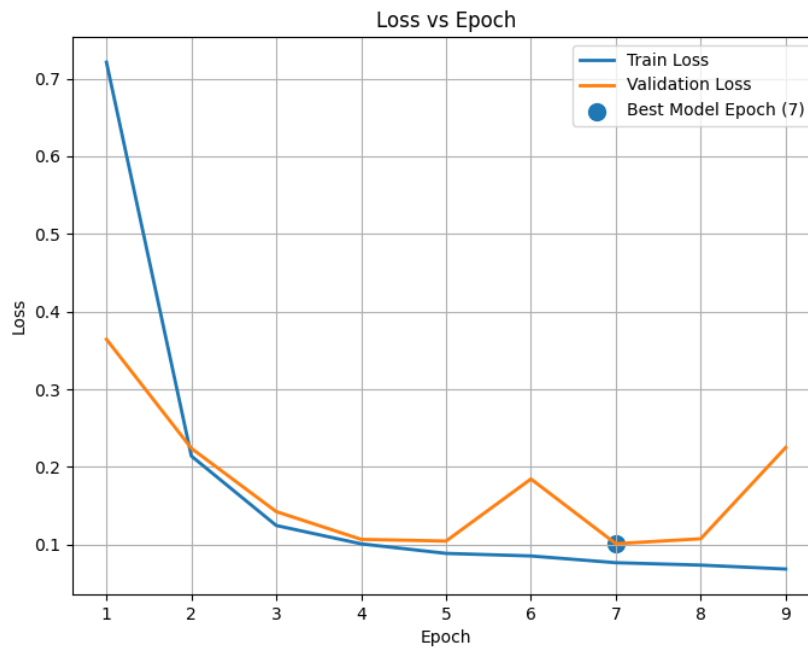


Figure 1: Training and Validation Loss vs Epoch

7.2 Dice Score vs Epoch

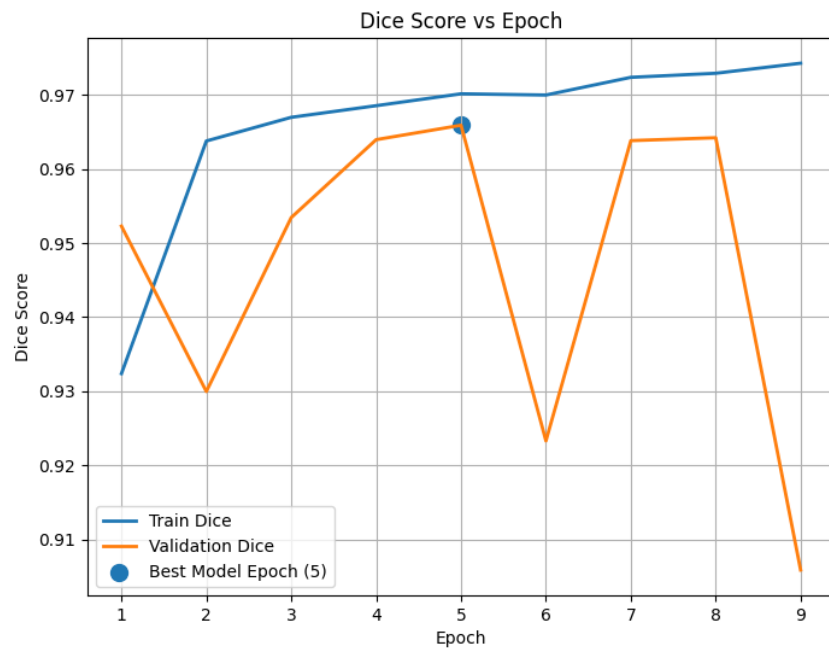


Figure 2: Dice Score vs Epoch

7.3 IoU vs Epoch

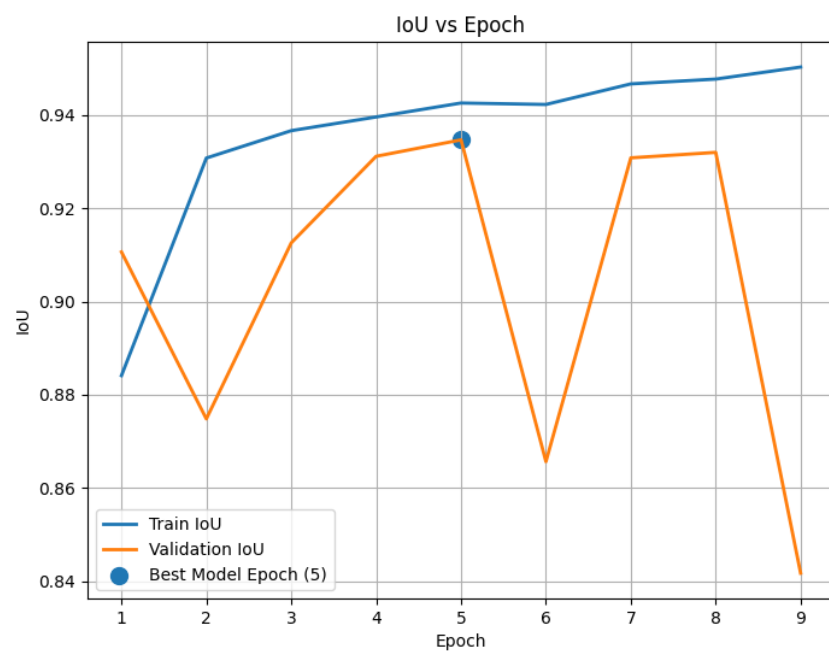


Figure 3: IoU vs Epoch

8 Segmentation Results

8.1 Qualitative Results

The following figure demonstrates the segmentation performance of the proposed model.

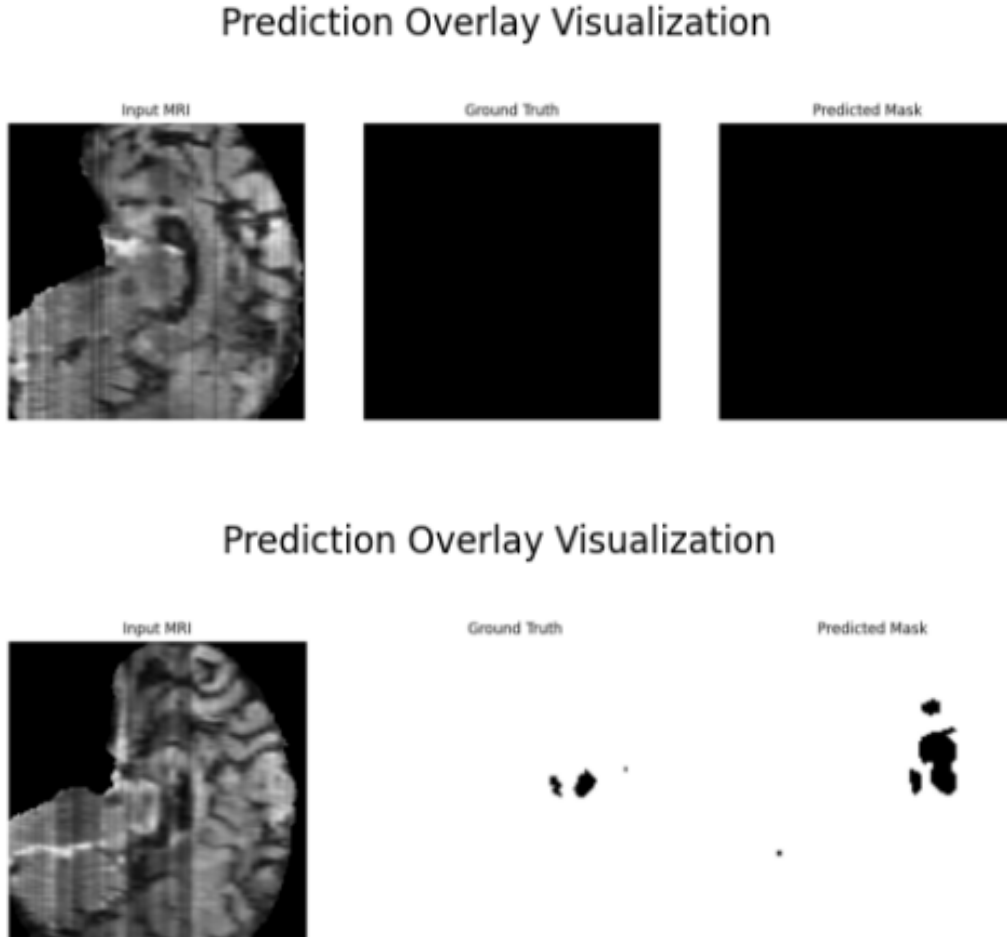


Figure 4: Input MRI, Ground Truth, and Predicted Segmentation

The predicted masks closely match the ground truth segmentation regions, indicating strong tumor localization capability.

9 Explainability using Grad-CAM

Grad-CAM visualizations were generated to analyze the regions focused on by the model during segmentation.

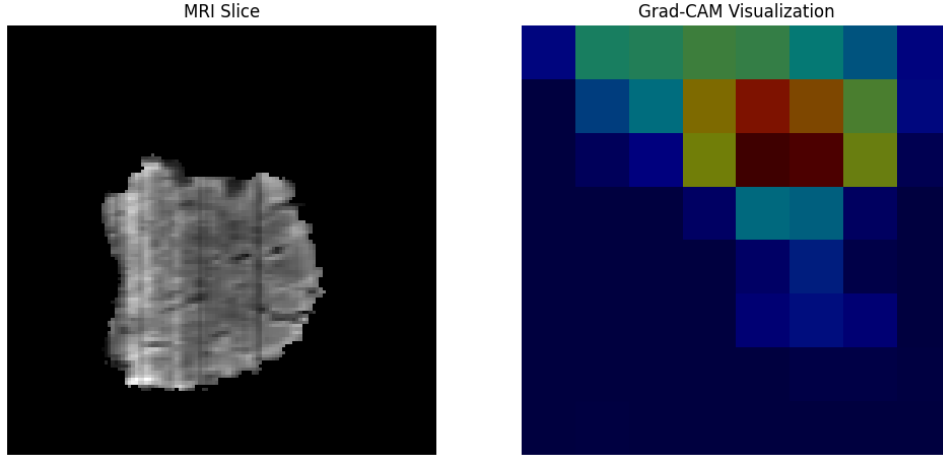


Figure 5: Grad-CAM Visualization

The heatmaps indicate that the model focuses on tumor-relevant regions within the MRI scans.

10 Embedding Analysis using t-SNE

A t-SNE visualization was generated to analyze the alignment between image and text embeddings.

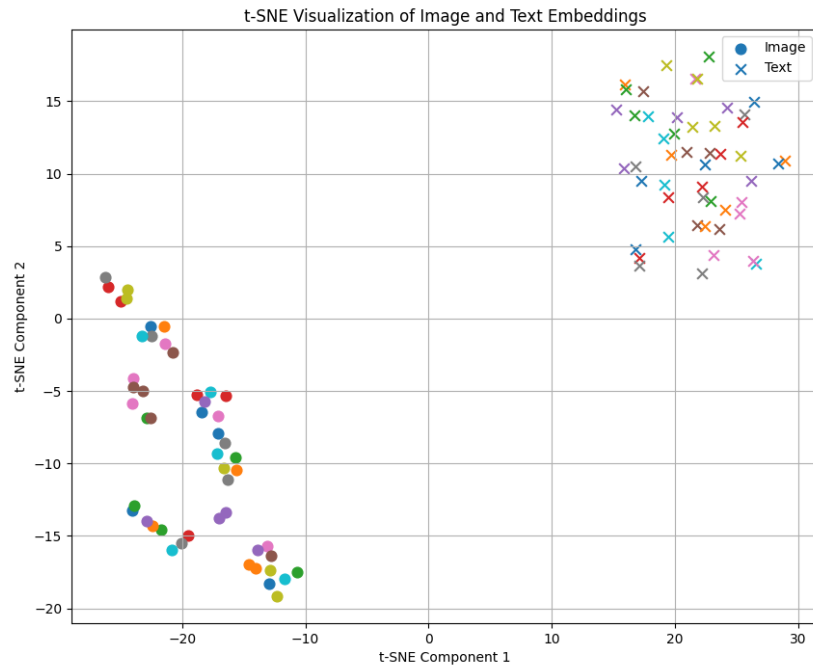


Figure 6: t-SNE Visualization of Image and Text Embeddings

Distinct clustering patterns demonstrate that the multimodal framework successfully learned structured latent representations for both MRI and textual modalities.

11 Error Analysis

11.1 Dice Score Distribution

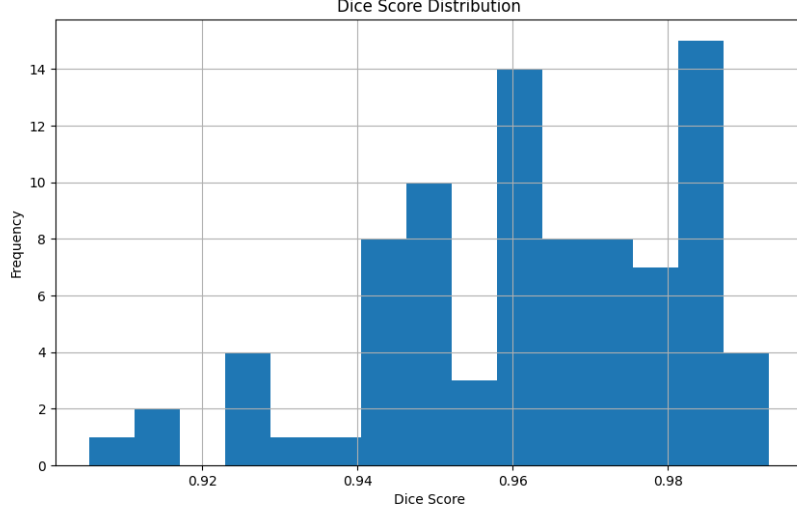


Figure 7: Dice Score Distribution

The Dice score distribution indicates consistent segmentation performance across validation samples.

11.2 Failure Case Analysis

The proposed framework achieved zero severe failure cases below a Dice threshold of 0.90, indicating robust segmentation performance.

12 Discussion

The multimodal approach demonstrated substantial improvement in segmentation quality by incorporating textual guidance into the segmentation pipeline.

Key observations include:

- Strong overlap between predicted and ground truth masks
- Stable validation performance across samples
- Effective multimodal embedding learning
- Robust feature localization demonstrated through Grad-CAM

The integration of textual descriptions enabled semantic enhancement of spatial feature learning.

13 Conclusion

This project proposed a multimodal brain tumor segmentation framework integrating MRI scans and radiology reports using Vision Language Models.

The framework successfully:

- Learned joint image-text representations
- Achieved high segmentation accuracy
- Demonstrated stable performance
- Produced interpretable visualizations

The experimental results validate the effectiveness of multimodal learning for medical image segmentation.

Future work may include:

- Transformer-based decoders
- Attention-guided fusion
- Multi-modal MRI integration
- Real-time clinical deployment